Q2 Training by Graphical Illustration

$$E(w) = w_1^2 - 2w_1 + 1 + 2(w_2^2 - 4w_2 + 4)$$

suppose $w = [0.5 \quad 1]^T$

a) Batch Descent grad. w/ Contour Plots.

Complete square → min. at $w = [1, 2]^T$

$$E(w) = w_1^2 - 2w_1 + 1 + 2(w_2^2 - 4w_2 + 4)$$

$$= (w_1 - 1)^2 + 2(w_2 - 2)^2$$



Set E = Constant c ⇒ $E = c$.

$c = (w_1 - 1)^2 + 2(w_2 - 2)^2$ Now find where c crosses line through origin / centre.

Horizontal line through centre ⇒ $w_2 = 0$

$c = (w_1 - 1)^2$

$w_1 = \sqrt{c} \pm 1$

⇒ contour is ellipse centred at $(1, 2)$

Vert. line through Centre ⇒ $w_1 = 0$

$c = 2(w_2 - 2)^2$

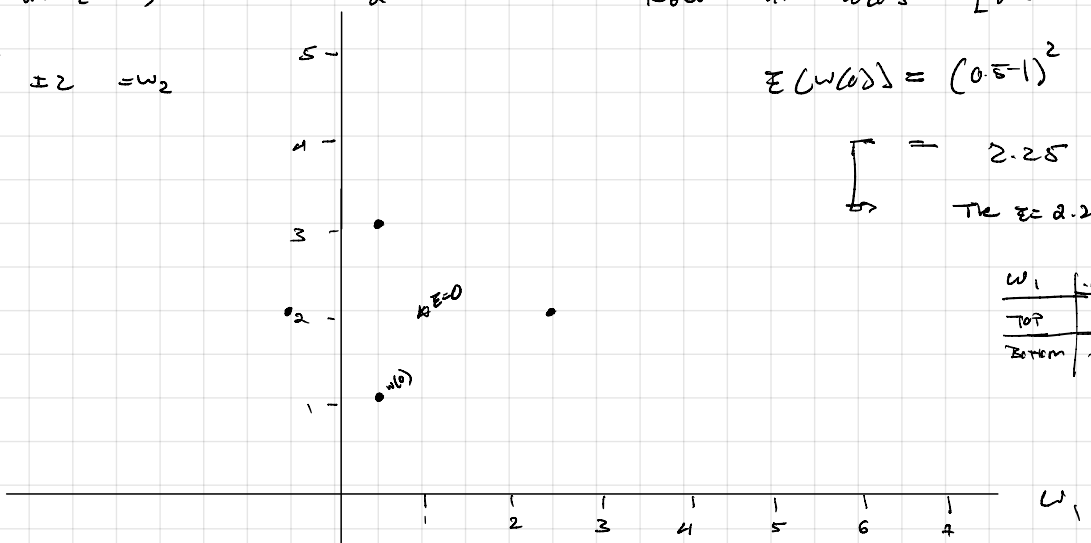
$\sqrt{\frac{c}{2}} \pm 2 = w_2$

Now at $w(0) = [0.5 \quad 1]^T$

$E(w(0)) = (0.5 - 1)^2 + 2(1 - 2)^2$

$= 2.25$

⇒ The $E = 2.25$ oval



w_1	-0.5	0	0.5	1	1.5	2.5
Top	2	2.79	3	3.06	3	2
Bottom	2	1.21	1	0.94	1	2

Suppose points are plotted ⇒ Now gradient.

Diff. w.r.t w_1 and w_2

$$E = w_1^2 - 2w_1 + 1 + 2w_2^2 - 8w_2 + 8$$

Try $w(0) = [0.5 \quad 1]$

$\frac{dE}{dw_1} = 2(0.5) - 2 = -1$

$\frac{dE}{dw_1} = 2w_1 - 2$ $\quad$ $\frac{dE}{dw_2} = 4w_2 - 8$

$\frac{dE}{dw_2} = 4(1) - 8 = -4$ ⇒ $\nabla E = [-1 \quad -4]$

↗ new.

h is just $-\nabla E$

then Line minimization

$h = -\nabla E = [1, 4]$ ⟹

- start at $w(0) = 0.5, 1$
- new point = old + η · direction

$= [0.5, 1] + \eta \cdot [1, 4]$

$w_1 = 0.5 + \eta$

$w_2 = 1 + 4\eta$ ⟹ Finding η

η	w_1	w_2	E
0	0.5	1	2.25
0.1	0.6	1.4	0.86
0.2	0.7	1.8	0.17
0.25	0.75	2	0.0625 *
0.3	0.8	2.2	0.12
0.4	.9	2.6	0.73
0.5	1	3	2

Chose $\eta \approx 0.25$

Now ⟹ $E = (0.5 + \eta - 1)^2 + 2(4\eta - 1)^2$

$E = 33\eta^2 - 14\eta + 2.25$

$\frac{dE}{d\eta} = 66\eta - 14$

$66\eta - 14 = 0$

$\eta = 14/66 \rightarrow 0.2546$

1 epoch ↓

Now $w(1)$

$w_1 = 0.5 + 0.2546$

$= 0.7546$

$w_2 = 1 + 4(0.2546)$

$= 2.0303$

$w(1) = [0.7546, 2.0303]^T$ ↓

$E = (0.7546 - 1)^2 + 2(2.0303 - 2)^2$

$= 0.0606$

↑ Draw

Now Epoch 2

$\frac{dE}{dw_1} = 2w_1 - 2$

$= 2(0.7546) - 2$

$= -0.4908$

$\frac{dE}{dw_2} = 4w_2 - 8$

$= 4(2.0303) - 8 = 0.1212$

$\nabla E = [-0.4908, 0.1212]$

$h = -\nabla E = [0.4908, -0.1212]$

new = old + η · dir

$= (0.7546, 2.0303) + \eta \cdot (0.4908, -0.1212)$

$w_1 = 0.7546 + 0.4908\eta$

$w_2 = 2.0303 - 0.1212\eta$

η	w_1	w_2	E
0	0.7546	2.0303	0.0606
0.2	0.8527	2.0061	0.0212
0.4	0.9515	1.9818	0.0030
0.5	1	1.9697	0.0012 *
0.6	1.0485	1.9576	0.0057
0.8	1.1454	1.9333	0.03

$E = (0.4908\eta - 0.2424)^2 + 2(0.0303 - 0.1212\eta)^2$

$E = 0.2644\eta^2 - 0.2424\eta + 0.0606$

$\frac{dE}{d\eta} = 0.5288\eta - 0.2424$

$\eta = 0.2424 / 0.5288 = 0.4582$ ⟹ $w(2) = [0.9855, 1.9431]$ ⟹ $E = 0.00153$